

Assignment 1

Q1 a) AI System:- IBM Deep Blue

* Core Objective:-

IBM's Deep Blue was developed to defeat a reigning world chess champion under standard tournament conditions. In 1997, it defeated Garry Kasparov, marking a historic milestone in AI.

* Theoretical Foundation:-

Deep Blue was built on:-

1) State Space Search:-

- Chess is modelled as a search problem.
- Each board configuration = state
- Legal moves = transitions
- Goal = checkmate.

2) Minimax Algorithm:-

- Assumes rational opponent
- Maximises its own utility while minimizing opponent's

3) Alpha Beta Pruning:-

- Eliminates branches that cannot affect final decision
- Dramatically improves efficiency.

4) Evaluation Function:-

- Assigns numerical values to board positions.
- Factors include material balance, King safety, mobility, pawn structure.

* Reflection of AI goals:-

Deep Blue reflects classical AI goals:-

- Rational decision making
- Problem solving
- Optimal move selection under constraints
- Simulation of intelligent strategic reasoning

It embodies the symbolic AI paradigm, focusing on logical reasoning rather than learning.

b) Architecture Diagram:-

1) User Interface Layer:-

This is where moves are presented to and from the system. In a match against a human, the UI would take the current board state and return the chosen best move.

2) Software Control Layer:-

- Runs on IBM RS/6000 SP supercomputer under IBM AIX OS.
- Executes the chess playing program written in C.
- Coordinates move search and delegates tasks to lower layers.

3) Search and Evaluation Layer:-

- Implements minimax with alpha-beta pruning search logic
- Evaluation function scores positions.
- Accesses opening books (grandmaster opening moves) and endgame databases to guide play.

4) Hardware layer:-

- General-purpose processors (about 30 PowerPC 604e CPUs) handle control and upper-level search tasks.
- Chess chips (480 custom-designed VLSI accelerators) massively speed up the search and evaluation of board positions in hardware.
- Together, they form a parallel system capable of evaluating $\sim 260M$ positions per second.

5) Memory and Storage:-

- Holds runtime data (positions, partial trees).
- Includes large chess databases (openings, endgame tables) to optimize play.

User Interface Layer
(Input chess position, Output
Best move / game decisions)

Software Control Layer

- Chess Program (C code)
- AIX operating system
- Search Algorithm controller

Search and Evaluation Layer

- Minimax / Alpha-Beta Search
- Evaluation function
- Opening book and Endgame DB

Hardware Layer

General-Purpose
Processors (30
PowerPC 604e)

← Bus →

Chess Chip
Accelerators
(480 VLSI ASIC)

Memory and Storage

- RAM (for position data)
- Databases (opening/endgame)

C) * Key Components:-

- 1) Specialized chess hardware processors.
- 2) Minimax search algorithm.
- 3) Alpha-Beta Pruning
- 4) Heuristic evaluation function
- 5) Opening book database
- 6) Endgame tablebases.

* Interaction Analysis:-

- Move generator feeds search engine.
- Search engine recursively explores tree.
- Evaluation function assigns heuristic scores.
- Hardware accelerates computation massively.
- Precomputed databases reduce search depth in known positions.

* Effectiveness:-

The interaction was extremely efficient for chess because:

- Chess has clearly defined rules.
- It is deterministic and fully observable.
- Evaluation criteria are well-defined.

However, intelligence was domain-restricted.

d) * Impact :-

- 1) Boosted AI research credibility
- 2) Demonstrated feasibility of brute-force + heuristics.
- 3) Inspired development of systems like AlphaGo by DeepMind.
- 4) Advanced parallel computing technologies.

* True vs Narrow Intelligence :-

Deep Blue represents narrow AI because:

- It operates only within chess.
- Cannot generalize learning.
- No reasoning outside predefined rules.

It does not exhibit:

- Self-awareness.
- Transfer learning.
- Autonomous learning capability.

Thus, it is highly specialized intelligence.

e) * Technical Limitations :-

- No learning ability.
- Hardcoded evaluation function.
- Requires enormous computational power.
- Not adaptable to new domains.

* Ethical Issues:-

- Minimal direct ethical risk.
- Raised philosophical debate: Is intelligence just computation?

* Practical Limitations:-

- Extremely expensive hardware.
- Not scalable to real world messy environments.

* Effect on Reliability:-

- Reliable within chess domain.
- Not reliable outside structured environments.

f) To make it more adaptive:

- 1) Integrate machine learning.
- 2) Use reinforcement learning (like AlphaZero).
- 3) Enable self-play adoption.
- 4) Improve explainability of decision making.
- 5) Develop transferable reasoning mechanisms.

A hybrid system combining symbolic reasoning and deep learning would enhance adaptability and social relevance.

Q2. Selected field = Healthcare.

a) A major emerging technology is deep learning for radiology diagnosis, used to detect diseases from X-rays, MRIs and CT scans.

Example systems:

- Google Health AI models.
- FDA-approved ~~diag~~ diagnostic AI systems.

b) * Working Mechanism:-

1) Data Collection:-

- Large dataset of labelled medical images.

2) Preprocessing:-

- Noise removal
- Normalization
- Augmentation

3) Deep Neural Networks:-

- Convolution Neural Networks (CNNs)
- Feature extraction layers.
- Classification layers

4) Training:-

- Backpropagation
- Loss function minimization
- Validation and tuning

5) Inference:-

- New image input
- Model predicts disease probability.

* Applications:-

- Cancer detection (lung, breast)
- Brain tumor identification.
- Diabetic retinopathy detection.
- Pneumonia detection from chest X-rays
- Stroke detection in CT scans.

AI systems can detect subtle patterns invisible to human eyes.

c) * Advantages:-

- 1) High diagnostic accuracy.
- 2) Faster detection.
- 3) Reduces doctor workload
- 4) Early disease detection
- 5) Scalable to remote areas.

* Challenges:-

- 1) Requires massive labeled datasets.
- 2) Data imbalance problems.
- 3) High computational cost.
- 4) Integration with hospital systems.

* Ethical Issues:-

- 1) Patient data privacy.
- 2) Algorithmic bias
- 3) Accountability in misdiagnosis
- 4) Over-reliance on automation

d) Future Improvements:-

- 1) Federated learning for privacy-preserving training.
- 2) Explainable AI to justify medical decisions.
- 3) Hybrid human-AI collaborative system.
- 4) Improved fairness through diverse datasets.
- 5) Real time multimodal analysis (combining imaging + patient history).

Q3 a)

1) * Performance Measure:-

- Accuracy of selecting qualified candidates.
- Reduction in hiring time.
- ~~Provision~~ Fairness and bias minimization.
- Precision and recall in shortlisting.
- Employer satisfaction.

* Environment:-

- Online job portals.
- Company HR databases.
- Resume repositories (PDF, DOC, LinkedIn profiles).
- Applicant tracking systems.
- Dynamic job market.

* Actuators:-

- Rank candidates.
- Shortlist or reject resumes.
- Send automated emails.
- Update HR dashboard.
- Flag suspicious or incomplete applications.

* Sensors:-

- Resume text (education, skills, experience).
- Job description requirements.
- Keywords and certifications.
- Candidate metadata (location, years of experience).
- Feedback from recruiters.

2) * Performance Measure:-

- Accuracy of flood prediction.
- Early warning time.
- False positive/negative rate.
- Coverage area monitored.
- Reduction in disaster damage.

* Environment:-

- Rivers and dams
- Urban drainage systems
- Weather systems
- Satellite monitoring systems.
- IoT sensor networks.

* Actuators:-

- Issue alerts to authorities
- Send SMS notifications
- Activate sirens.
- Trigger emergency protocols.
- Update disaster dashboards.

* Sensors:-

- Rainfall data
- River water level sensors.
- Soil moisture sensors
- Satellite images
- Weather forecasts
- Historical flood data.

Resume Screening System:

b) AI based

Environment	Property	Type	One - line Reasoning
Observability		Partially Observable	The system cannot know hidden candidate qualities like honesty or soft skills fully.
Single Agents		Single Agent	It operates independently without competing intelligent agents.
Deterministic vs Stochastic		Stochastic	Human decisions and future job performance are uncertain.
Episodic vs Sequential		Episodic	Each resume evaluation is independent of others.
Static vs Dynamic		Semi-Dynamic	Job requirements may change over time.
Discrete vs Continuous		Discrete	Resumes and decisions are presented as separate cases.
Known vs Unknown		Known	Rules and evaluation criteria are predefined

Environment	Property	Type	One-line Reasoning
Observability		Partially Observable	Not all environmental factors (underground water flow, future rainfall) are fully visible.
Agents		Single Agent	It monitors and predicts without competing intelligent agents.
Deterministic vs Stochastic		Stochastic	Weather and rainfall patterns are unpredictable.
Episode vs Sequential		Sequential	Current flood risk depends on previous rainfall and water levels.
Static vs Dynamic		Dynamic	Environmental conditions change continuously.
Discrete vs Continuous		Continuous	Water levels and rainfall vary continuously over time.
Known vs. Unknown		Partially Known	Physical laws are known but exact future events are uncertain.

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* Implicit Agent: Automated Book Retrieval System (robotic system).

* State Space Search:-

1) State Space:-

Each state can be defined as:

$$\text{state} = (x, y, \text{book-status})$$

where:

- (x, y) = current grid position of the robot
- book-status = 0 (not picked) or 1 (picked)

State space includes:-

- All accessible grid positions.
- Excludes blocked aisles.

2) Initial State:-

$$\text{initial state} = (x\text{-start}, y\text{-start})$$

It would be $(0, 0)$

3) Actions:-

From any valid position, the robot can:

- Move up
- Move down
- Move left
- Move right

Action constraints:-

- Cannot move outside grid.
- Cannot move into blocked aisle.
- Cannot pass through walls/shelves.

4) Transition Model:-

If the current state is (x, y) then:

- Move up $\rightarrow (x-1, y)$
- Move down $\rightarrow (x+1, y)$
- Move left $\rightarrow (x, y-1)$
- Move right $\rightarrow (x, y+1)$

The resulting cell should be valid and not blocked. If movement leads to a blocked aisle then the state does not change.

5) Goal Test:-

The goal is satisfied when:

current position == book shelf position

When the robot reaches that location, it picks the book and stops.

6) Path Cost:-

Cost of each move = 1 unit

Path cost = sum of all movement costs

7) Solution:

A solution is a sequence of valid moves that takes the robot from the initial state to the goal state with minimum cost.

Qs. 1. The drone is using Iterative Deepening Search.

It explores the search tree up to a fixed depth limit. If the exit is not found, it;

- Returns to the start
- Increases the depth limit
- Repeats the search.

This continues until the goal is found. It combines:

- The memory efficiency of DFS
- The completeness of BFS.

2. a) Limited Memory:-

- The drone cannot store a large search tree
- BFS requires storing all nodes at a level (high level memory)
- DFS requires very little memory

IDS behaves like DFS in memory usage:

- It only stores nodes along one path at a time
- Memory usage is small.

So IDS is suitable.

b) Unknown Depth of Exit:-

- The drone doesnot know how deep the exit is.
- If it uses DFS, it may go very deep into a wrong tunnel.
- If it uses BFS, memory may explode.

IDS solves this:

- It gradually increases search depth
 - It guarantees finding the shallowest goal.
 - It doesnot assume any depth in advance
- So it works well when goal depth is unknown.

c) Very deep dead-end tunnels:-

If DFS is used:

- It may go extremely deep into a dead end
- Waste time exploring useless paths.

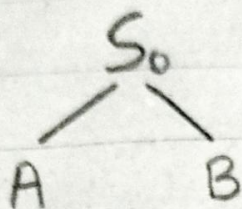
IDS prevents this:

- It limits depth each time.
 - It avoids getting stuck forever in deep dead ends.
 - It eventually explores all shallow paths first.
- So IDS protects against infinite or very deep branches.

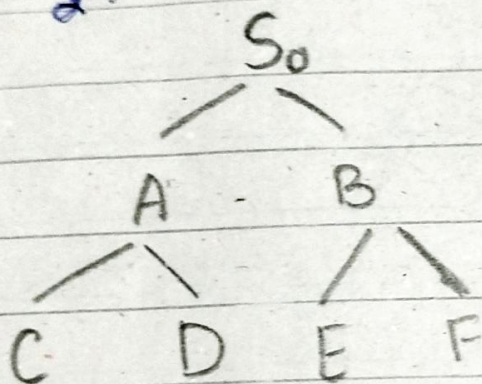
3. Depth 0:

S_0

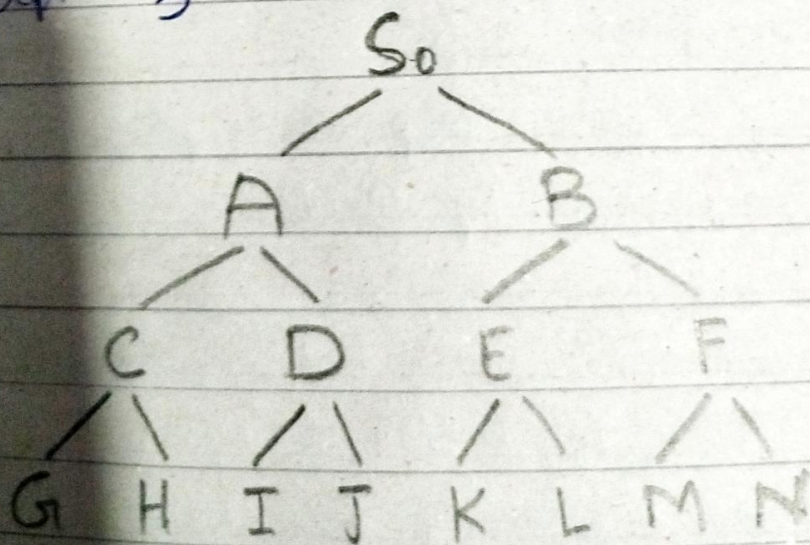
Depth 1:



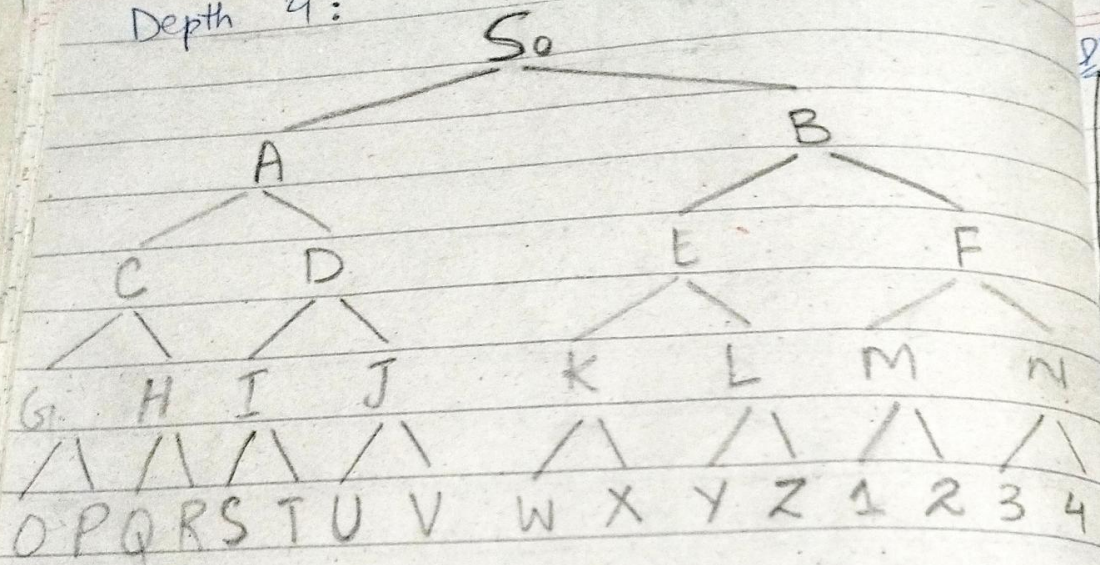
Depth 2:



Depth 3:



Depth 21:



Q6 3. In this maze:

- Goal depth is unknown.
- There are dead ends.
- DFS may explore deep wrong path first.
- IDS explores level by level.

So:

DFS with depth limit works but depends on correct limit. IDS is better because:

- It does not require knowing goal depth.
- It guarantees finding shortest path.
- It avoids getting stuck in deep dead ends.
- It uses memory efficiently.

Therefore, Iterative Deepening Search is more suitable for this maze problem than Depth First Search.